The Association Between the Kaitz Index and Employment in Puerto Rico

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Presentation Overview

Research Context

- **Objective:** Examine the relationship between the Kaitz Index and employment levels in Puerto Rico.
- **Data:** Panel data at the postal code (ZIP) level from Q1 2012 to Q4 2023.
- **Key Gap:** Exploring the **spatial dynamics**—how wage conditions in one region affect neighboring areas.
- **Methodology:** Spatial Durbin Model (SDM) with Fixed Effects.

Literature Summary

- **Negative Relationship:** Freeman (1992) found 8%–10% employment reduction; Neumark (2000) linked high minimum wages to high unemployment.
- **Mixed Results:** Card & Krueger (1994) found no significant effect; Caraballo-Cueto (2016) observed slight positive short-run effects.
- **Puerto Rico Specific:** Federal Reserve (2012) suggested subminimum wages for youth; Hernández (2017) found total employment reductions post-increase.

Defining the Kaitz Index

The Kaitz Index (K) represents the ratio of the minimum wage to the average wage:

$$K_{it} = \frac{m_t}{\bar{w}_{it}}$$

Variables

- m_t: Minimum wage at time t.
- $\bar{w}_{it}$: Average wage in zip code i at time t.

Source: FRED and QCEW (Department of Labor).

Spatial Durbin Model (SDM)

To account for spatial dependence, we use:

$$Y_{it} = \alpha + \beta_1 X_{it} + \rho W Y_{it} + \gamma W X_{it} + \epsilon_{it}$$

- Y_{it}: Employment in zip code i.
- W: Spatial weights matrix (Adjacency).
- ρ: Spatial autoregressive parameter.
- $\gamma W X_{it}$: Spatial lags of independent variables (Indirect effects).

Spatial Weighting Strategies (W)

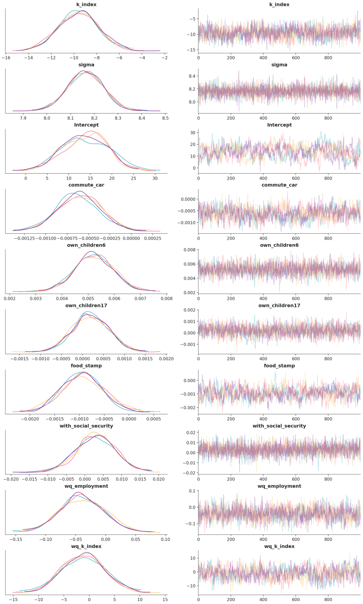
Four distinct weights are implemented for robustness:

1. W_{queen}: Contiguous borders.
2. W_{knn6}: 6 nearest neighboring regions.
3. $W_{distance}$: 30-mile radius.
4. W_{tensor}: Semi-parametric diagnostic using centroids to check for non-uniform spatial correlation.

Queen Weights: Parameter Estimates

Variable	Mean	3% CI (L)	3% CI (U)	R-Hat
Kaitz Index	-9.594	-12.806	-6.093	1.00
Sigma	8.164	8.013	8.300	1.00
Intercept	13.808	3.715	23.321	1.01
Commute by Car	-0.001	-0.001	-0.000	1.01
WQ Employment	-0.040	-0.106	0.036	1.00
WQ K Index	-1.139	-8.625	7.089	1.00

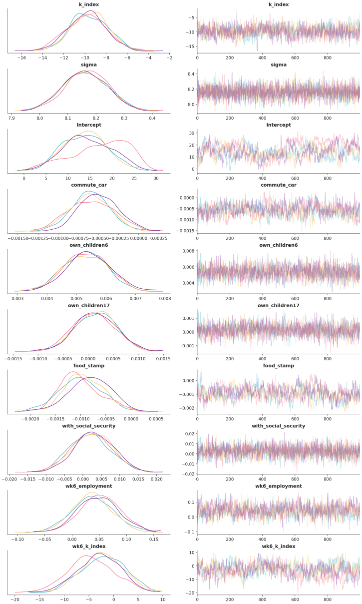
Queen Weights: Posterior Distribution



KNN6 Weights: Parameter Estimates

Variable	Mean	3% CI (L)	3% CI (U)	R-Hat
Kaitz Index	-9.687	-12.917	-6.293	1.02
Sigma	8.163	8.015	8.307	1.00
Intercept	14.758	4.748	25.399	1.11
WK6 Employment	0.044	-0.031	0.113	1.02
WK6 K Index	-2.916	-11.209	5.525	1.08

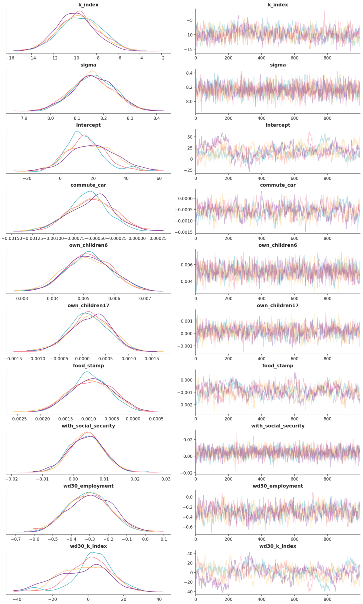
KNN6 Weights: Posterior Distribution



Distance Weights: Parameter Estimates

Variable	Mean	3% CI (L)	3% CI (U)	R-Hat
Kaitz Index	-9.650	-13.154	-6.244	1.03
Sigma	8.160	8.009	8.307	1.00
Intercept	17.368	-5.015	45.195	1.03
WD30 Employment	-0.301	-0.519	-0.090	1.01
WD30 K Index	-0.084	-28.740	23.926	1.04

Distance Weights: Posterior Distribution



Tensor Diagnostic Plot

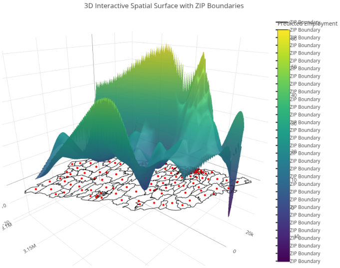


Figure: Tensor Product Spatial Smoothing Diagnostic

- Used to determine if spatial correlation is uniform.
- Results suggest non-uniform spatial relationships across centroids.

Key Findings & Future Work

- **Employment Impact:** Confirmed negative and significant relationship between Kaitz Index and employment in PR.
- **Spillovers:** Evidence of negative spillover effects on neighboring ZIP codes, though MCMC convergence makes magnitude precise estimation difficult.
- **Future Research:**
  - Explore **lagged spatial effects**.
  - Utilize more complex spatial smoothing given the non-uniform nature of regional relationships.

Thank You!